

B.Tech. CSE (III YEAR - VI SEM) (2025-2026)

DEPARTMENT OF COMPUTER ENGINEERING & APPLICATIONS



GLA University

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Project Title:

**Clinician Unchained – Agentic Workflow Engine for Prescription
Resolution**

Team Member 1:

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Mentor:

Prof. Yunis Lone

Project Synopsis: Clinicians Unchained – Agentic Workflow Engine for Prescription Resolution

0. Cover

- **Project title:** Clinicians Unchained – Autonomous AI System for Prescription Resolution
- **Team name & ID:** Team Neon(T-88)
- **Institute / Course:**GLA University (MATHURA), B-Tech CSE (AIML & IOT)
- **Version:** v1.0
- **Date:**

Revision history

Version	Date	Author	Change
v0.1	15 Aug 2025	Rishabh Sengar	Initial draft
v1.0	26 Aug 2025	Team Neon	Filled with examples

1. Overview

- **Problem statement:** Clinicians spend a large amount of time resolving prescription-related administrative issues such as insurance denials, prior authorizations, and pharmacy stock problems. These issues require repeated phone calls and manual follow-ups, leading to delayed patient care and increased clinician burnout.
 - **Goal:** To build an AI-driven agentic workflow system that autonomously resolves prescription issues by communicating with insurance providers and pharmacies without clinician involvement.
 - **Non-goals:** Direct Payments, Real hospital system integration and Production – level deployment.
 - **Value proposition:** Automates repetitive healthcare administrative tasks using AI agents, saving clinician time and improving patient outcomes.
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2. Scope and Control

2.1 In-scope

- Automated prescription issue detection
- AI reasoning for rejection analysis
- Simulated AI voice agent workflows
- Decision-making automation
- Notification system for clinicians
- Logging and audit tracking

2.2 Out-of-scope

- Real telecom infrastructure
- Live insurance databases
- Full hospital EMR integration

2.3 Assumptions

- Prescription data is provided digitally
- Simulated calling APIs represent real communication
- Pilot system for demonstration

2.4 Constraints

- Limited development time
- Student-level infrastructure
- Prototype deployment

2.5 Dependencies

- AI language model API
- Workflow automation tool (n8n)
- Email notification service

2.6 Acceptance criteria and sign-off

- System detects prescription failure
- AI identifies issue reason
- Workflow executes corrective steps
- Clinician receives final status

- **Sign-off:** Mentor approves demo and test report; all P1 bugs closed.

Stakeholder	Role	Decision area	Signature/Approval	Date
Prof. Yunis Lone	Mentor	Scope, final acceptance		
Rishabh Sengar	Product Lead	Release readiness		

3. Stakeholders and RACI

Activity	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Requirements	Rishabh	Rishabh	Mentor	Team
Design	Ayaan	Ayaan	Mentor	Team
Implementation	Ayaan	Ayaan	—	Team
Testing	Kartikay	Kartikay	Mentor	Team
Release	Rishabh	Mentor	Mentor	Dept

4. Team and Roles

Member	Role	Responsibilities	Key skills	Availability	Contact
Rishabh Sengar	Team Leader	Architecture, AI logic, workflow design	AI, Automation	10 hrs/wk	rishabh.sengar_cs.aiml23@gla.ac.in
Mohd Ayaan Hussain	Tech Lead & Developer	Workflow implementation, testing	APIs, Logic	8 hrs/wk	Mohd.hussain_cs.aiml23@gla.ac.in
Karika y Garg	Developer	UI, API & documentation	Frontend, Docs	8 hrs/wk	Kartikay.garg_cs.aiml23@gla.ac.in

5. Week-wise Plan and Assignments

Week	Milestone	Deliverables
1	Requirements freeze	Problem Analysis
2	System design	Architecture diagram
3	AI logic	Reasoning engine
4	Workflow Automation	n8n flows
5	Simulation calls	Mock communication
6	Notifications	Email system
7	Testing	Debugging
8	Final demo	Report & presentation

6. Users and UX

6.1 Personas

- **Clinician Alex:** Wants minimal administrative interruptions
- **Patient Sam:** Wants faster medication access

6.2 Top user journeys

Prescription rejected → AI analyze issue → AI communicates with insurer/pharmacy → AI resolves or escalates → Clinician notified

6.3 User stories

- As a clinician, I want prescription issues handled automatically so I can focus on patients.
- As a system, I want to identify rejection causes and act without human help.

6.4 Accessibility & localization

- Keyboard-only navigation on core flows; contrast AA; language: English; RTL not required.
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7. Market and Competitors

7.1 Competitor table

Competitor	Product	Strength	Weakness	Our Advantage
Manual admin	Human calls	Reliable	Slow	Automation
Call centers	Outsourced	Available	Costly	AI-based
Basic bots	Chatbots	Fast	Limited	Smart agents

7.2 Positioning

- **Unique angle:** Trust and speed within a single campus.
 - **Measurable delta:** Median listing time ≤ 90 s vs 3–5 min on general sites.
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8. Objectives and Success Metrics

- Reduce resolution time by 70%
 - Eliminate manual calls
 - Improve clinician efficiency
 - High task completion rate
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9. Key Features

Feature	Description	Priority
AI Decision Engine	Analyzez prescription failures	Must
Autonomous Agents	Simulated calling workflows	Must
Workflow Automation	Task execution	Must
Notification System	Alerts clinicians	Must
Logs	Action tracking	Should

10. Architecture

High-level

- Frontend Dashboard
- AI Reasoning Engine
- Workflow Automation (n8n)
- Mock Communication APIs
- Notification Service

11. Data Design

11.1 Data dictionary

Entity	Field	Type	Null ?	Allowed values	Source	Notes
Prescription	id	UUID	No	—	System	PK

Prescription	Medication_name	String	No	RFC 5322	Clinicain	Unique
Prescription	Status	String	No	—	System	PK
Prescription	Rejection_Reason	String(120)	Yes	—	Insurance/Pharmacy	Indexed
Agent_Action	Id	UUID	No	≥ 0	Seller	—
Agent_Action	Action_type	String	No	Call insurance, Call pharmacy	AI Agent	
Resolution	id	UUID	No	--	System	PK

11.2 Schemas and migrations

- ER Diagram included in Appendix
- Schema versioning maintained
- Rollback tested in staging environment

12. Technical Workflow Diagram

- State Transition Diagram
- Sequence Diagram
- Data Flow Diagram
- Work Architecture Diagram

13. Quality: NFRs and Testing

Functional

- Workflow correctness
- AI decision accuracy

Non-functional

- Reliability
 - Performance
 - Error handling
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14. Security and Compliance

14.1 Threat model (STRIDE)

- Basic authentication
- Data access control
- Secure API handling

14.2 AuthN/AuthZ

- Authentication through secure login using email and password
- Role-based access control with roles such as clinician, AI agent, and admin
- Authorization checks implemented at workflow and API level

14.3 Audit and Logging

- Logs all prescription status changes
- Records AI agent actions and decisions
- Tracks workflow execution history
- Maintains logs for monitoring and debugging purposes

14.4 Compliance

- Academic prototype following institutional guidelines
 - No real patient data used
 - Simulated healthcare workflows for demonstration
 - Data handled securely within project scope
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15. Delivery and Operations

- Local deployment
- Demo-based evaluation

16. Risks and Mitigations

16.1 Risk heatmap

Risk	Probability	Impact	Mitigation
AI errors	Medium	Medium	Manual override
Workflow failure	Medium	Medium	Retry Logic
Time constraints	Low	High	Simplified scope

17. Research and Evaluation

- Measure time saved
- Analyze success rate
- User feedback

18. Appendices

- **Glossary:** AI Agent,Workflow Automation, Prescription Denial, AuthN/AuthZ,Prototype.
- **References:** Course handbook; n8n docs; Youtube; claude code.